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Thesis Work 1

Development and Validation of a Cellular Automaton Cancer Model

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Abstract

In this work, I focus on further developing and validating a cellular automaton model for tumor growth that I built during my Tutored Research Project based on a previously available model. The project aims to simulate the proliferation and growth of cancer as well as the response of the immune system, and provide an easy to use interface and deep customizability alongside simple visualization and validation techniques.

While during the last year I mostly directed my time towards coding new features, like machine-learning elements or a Streamlit dashboard, and expanding the model itself as well as its python package TCAMpy, in the previous months I concentrated on validation, whose results I describe here. I wanted to see how similarly my model behaves to real tumors in terms of growth dynamics, shape, and defense against the host. I analyzed real data from numerous articles, compared it to many simulations, and made modifications based on the results.

I created new plots and statistical descriptions about simulations, like a visualization of the tumor border at multiple time points, radius and area growth line charts, and circularity measures. With the help of these statistics, I came to the conclusion that many aspects of the model are very realistic: mostly linear radius increase over time, proliferation suppressed to the border due to contact inhibition and roughly circular shape with irregular border.

This project was incredibly important to see how faithful the model was to real behavior and provide guidance for future work. The python package and the Streamlit dashboard allow easy access to simulations for everyone and will continue to be updated to reflect the latest advancements of the model.

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1. Introduction

1.1. Cellular automata

Cellular automata (CA) are mathematical models used to study complex systems through simple, local rules. They are especially suitable for cancer models, as they allow us to see emergent, large-scale behaviour arising from the individual interactions between tumor cells on a grid. With the help of a cellular automaton, we can study the migration and proliferation of the cells, growth dynamics, and even host response, such as the immune system.

Time progresses in discrete steps, and at each step, every cell updates its state simultaneously based on a set of rules that typically depend only on its own current state and the states of its neighboring cells. In tumor simulations, these rules resemble the possibilities for actual cell actions: apoptosis, proliferation, migration, etc.

Cellular automaton models are frequently very (or over-) simplified versions of complex systems, and another disadvantage is their lattice dependence. Despite all of this, they are often very realistic and can be used to study these systems extremely well. Here, I show how strongly my version resembles real tumor growth while being computationally efficient.

1.2. The model and TCAMpy

The model and its python module are detailed in my Tutored Research Project, available on the following link: *TRP* or in the module *documentation*. (Changes made since the TRP work are described below.) Here I give a shorter summary about the model.

On a 2 dimensional square grid each spot can either be empty, or filled with 2 types of cancer cells or an immune cell. In each cycle, which is an hour by default, every existing cell chooses and completes an action based on probabilities: cancer cells can die from apoptosis, migrate to a free neighbor or proliferate to a free neighbor, with the probabilities either defined directly ($PA=0.01$ for apoptosis) or calculated from other parameters ($24 \cdot dt/CCT$) for proliferation). Stem tumor cells (STCs) do not apoptose ($PA=0$), and can divide indefinitely, creating either a regular tumor cell (RTC) with asymmetric division or another STC with symmetric division (with a probability of PS). RTCs can die, and they only divide up to a maximum number of times: the $pmax$ parameter. If a tumor cell does not apoptose, migrate or proliferate, it stays quiescent. (Migration and proliferation are not possible if there are no free neighbors) [1].

Immune cells spawn around the tumor in a "frame" that grows with the colony, keeping the spawnpoints at a certain distance. They move towards tumor density, and they can take multiple steps per cycle, since they are much faster than tumor cells. When they come into contact with a target (stepping onto a neighboring spot), they might kill it with a small chance. The effect of immune exhaustion (IE), a time-dependent decline in function, is also simulated. Equations for spawn and kill probabilities, movement, and IE are presented in the TRP work.

TCAMpy is the python module developed for this project. It allows the user to run simulations with custom parameters, visualize the growth with multiple plots, as well as an animation, and inspect many statistical properties of the results, like tumor size/confluence, radial front speed, circularity, or immune behavior. The usage of this module is detailed in its *documentation*, and installation is available from *PyPI*. On *Streamlit Community Cloud*, it can be used without any coding, thanks to its user-friendly graphical interface. Below is an example result after running a simulation with TCAMpy. It shows growth, mutation, circularity data, alongside an optional animation. When running multiple animations, averages with standard deviation are presented, where applicable (line and bar charts).

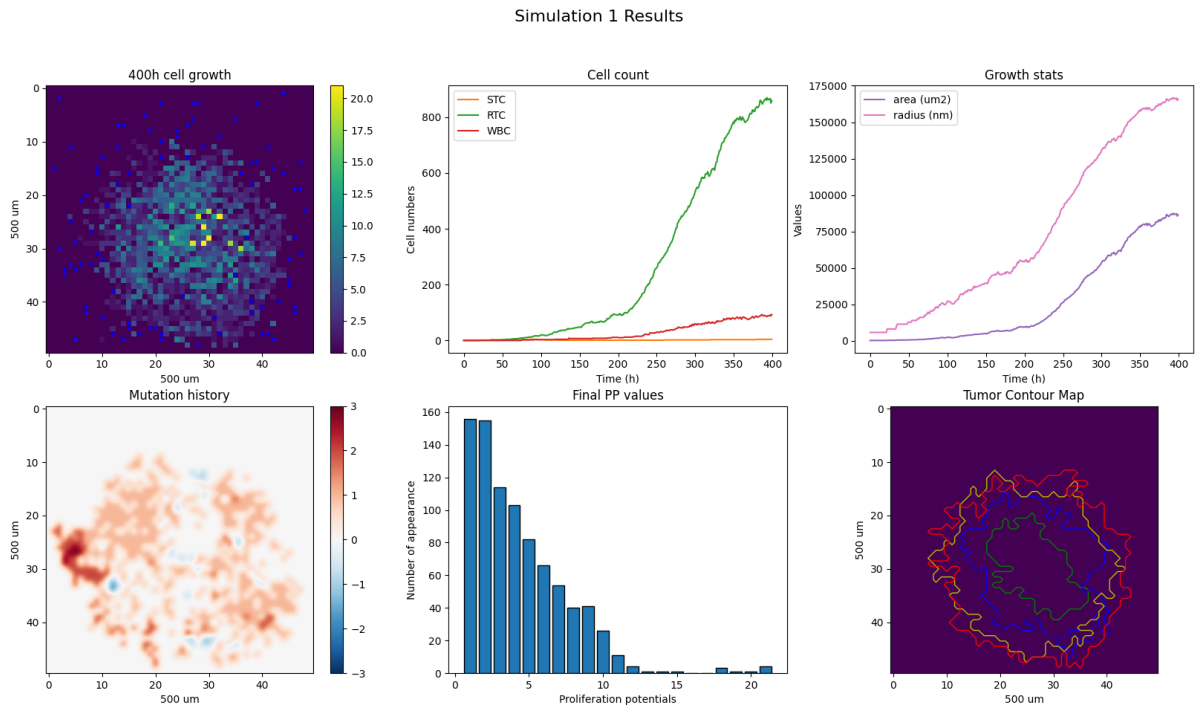


Figure 1.1: Result of a single simulation. Top left: tumor heatmap, top middle: cell numbers over time, top right: area and radius over time. Bottom left: mutation heatmap, bottom middle: proliferation potential values at the end, bottom left: tumor boundaries during growth.

1.3. Model parameters

Below you can see the list of the parameters of the model with their default values. The I parameter decides how strong immune cells are: their spawn rate, kill chance, lifespan and maximum motility, but other parameters of the immune response (that are not about the cells themselves, not calculated by I) are also listed here. These are last 4, and they currently cannot be changed upon creating a simulation with TCAMpy. (At least not in an easy way.)

Parameter	Datatype	Description	Default Value
cycle	int	duration of the model given in hours	500
side	int	the length of the side of field (10um)	50
pmax	int	maximum proliferation potential of RTC	20
PA	int	chance for apoptosis of RTC (in percent)	1
CCT	int	cell cycle time of cells given in hours	24
Dt	float	time step of the model given in days	1/24
PS	int	STC-STC division chance (in percent)	25
mu	int	migration capacity of cancer cells	4
ad	int	strength of cell-cell adhesion	0
I	int	strength of the immune cells (1-5)	3
M	int	tumor mutation chance (in percent)	10
offset	int	distance of spawnpoints ("frame") from the tumor	20
alpha	float	controls strength of immune (population) exhaustion	0.002
it_targ	float	desired mean immune/tumor ratio during simulation	0.1
infil	float	threshold for what cell counts as infiltrating (0-1)*	0.3

Table 1.1: Model parameters. *based on chemokine gradient, only for statistics.

1.4. Parameter influence

Machine learning features are also part of the model. Creating a large data set with randomized parameters, training a regression model (*random forest*), and predicting final tumor size or confluence for a new parameter set are possible, with more options to come in the future (such as more algorithms or aspects to predict). These features are available currently with both the installed TCAMpy module and the Streamlit dashboard as well.

Regression also helps identify which parameters have the most influence on the colony growth. Based on 1000 randomized simulations, *PS* came in first, with *CCT* being a very close second, not surprisingly, as both strongly affect proliferation speed or capacity. Adhesion strength in the third place means a strong change in contact inhibition at different *ad* values. *I* is also among the top influencing parameters, as the immune response can be the decision maker between survival and extinction. Other factors, such as *PA* had less significant effects, possibly due to the randomization range being not too big, as apoptosis is generally rare for RTCs (and impossible for STCs). These ranges were chosen to reflect viable values for different tumor types, like *CCT* ranging from 12 to 48 hours and *PS* from 20 to 75 percent, however, more studies need to be done to better identify the lower and upper limits for generating these datasets. (The exact ones used here, alongside code examples, can be seen in the documentation.)

The model was trained on 800 and tested on 200 simulations and had a r^2 score of 0.741 (and a MAE of 430.69), which means that it explains 74.1% of the variance in final tumor size across all runs. This might seem slightly low, but for a stochastic model with much randomness and possibility for radically different outcomes (even with identical parameters) and extinction, it can be considered relatively high, suggesting a real signal beyond noise.

2. Literature Review

2.1. Tumor growth dynamics

In order to be able to validate the model, I looked for a number of articles about real tumor-related data: growth curves, 2 dimensional shapes and images, proliferation and migration rules, etc. Here, I list my main findings based on publications, which I used to see how well the model simulated cancer growth. These also motivated the implementation of new options, statistics, and plots in the python module, which allows the user to closely inspect these properties after a simulation.

One of the most important points was that the increase in the colony radius is mostly linear, after a short exponential phase at the beginning [2]. This, by itself, could mean that not all cells proliferate in the tumor (at least not per cycle). Cell division is constrained to the border, while in the middle, the proliferation capacity is significantly lower due to contact inhibition: there is simply not enough space [3]. The doubling time is much longer than cell cycle times (can even be 100 fold in breast carcinomas) [2]. These findings closely match MBE dynamics (Molecular Beam Epitaxy), which says that growth share specific features, like linear radius increase, proliferation at the border, and cell diffusion at the edge leading to irregular, fractal-like boundaries. The movement can be understood as a search for space by the cells to avoid the mechanical response of the host tissue. However, it is not accepted today that universally all tumors belong to MBE dynamics. The boundary irregularity of cancers is mentioned multiple times in the articles, it is also an important aspect to look at in the model. Even though the edges are not smooth, the shape of the tumors remains roughly circular, as seen in figure 2.1 [2] [4] [5].

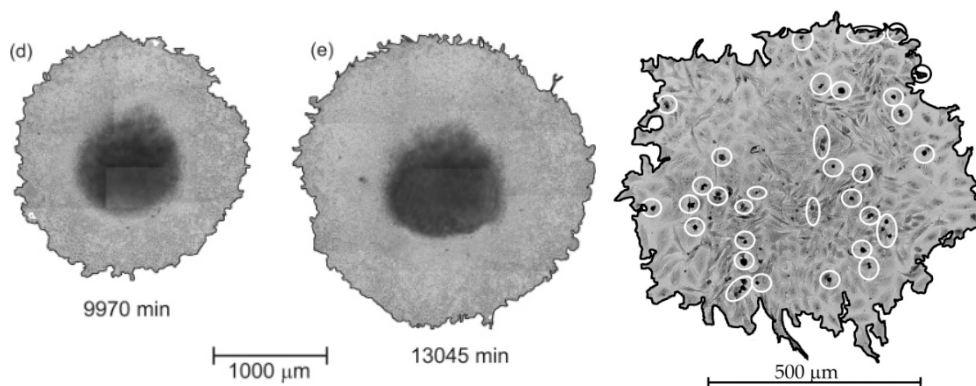


Figure 2.1: Radially growing HeLa colonies images showcasing border irregularity. Front velocity was reported to be around $0.9 \mu\text{m}/\text{min}$ [5].

2.2. Cell-cell adhesion

Cell-cell adhesion is a very important and dynamic mechanism reported in multiple articles [6], which can lead to phenomena like metastasis or collective migration. The latter is when clusters of cancer cells move together, because they are strongly connected to each other, and this connection is maintained through motion. However, cell-cell adhesion is not the primary force of the proliferation, migration and growth of tumor mass, whose more direct regulators are crowding and contact inhibition, but it can influence these parameters.

We can see that increased cell-cell adhesion strength decreases overall cell migration, but this effect is very complex in cancers, often leading to collective migration [7]. To avoid overcomplicating the model and keep it from getting computationally too expensive, I have decided to use a global adhesion strength parameter (seen above in table 1.1) that generally affects the migration of the cells, but this approach should and will be revisited later. (Mapping of individual cell contacts of neighboring target spots for migration and proliferation in order to modify target selection based on adhesion would be an expensive process.)

2.3. Immune response

I also wanted to check whether the immune cells respond realistically to the tumor growth, and modify the design of the response based on the results. I detail many important aspects of an immune attack against cancer cells in my previous Tutored Research Project (available on the link above or at the documentation), where I analyzed many immune-tumor articles. Here is a quick summary about the conclusions.

Immune cells migrate through tissue, forming motile kinapses with targets, and the per-cell killing capacity of immune cells in vivo is modest (2 to sometimes 16 kills per day) [8] [9] [10]. Killing by CTLs often requires multiple hits and single CTL–tumor cell contacts frequently fail. Lag time can be hours between contact and target death [10] [11], and tumor mutations reduce kill probability [12]. Immune motility (random walk + chemotaxis) in tissue varies (1–20 $\mu\text{m}/\text{m}$) and is reduced in dense tumor microenvironments, but is much higher than tumor motility [13] [14]. Immune recruitment is influenced by tumor size, chemokine gradients, and immune-strength; infiltration is often lower in “cold” tumors and higher in “hot” tumors [10]. Finally, there is also the effect of immune exhaustion: a time-dependent decline in functional cytotoxicity, meaning even for the same immune-cell count, kill efficacy may drop over time [15] [16].

3. Model Validation

3.1. Radius, area and front velocity

The first aspect I wanted to look at when validating the model was the growth of the colony. As discussed previously, the expected radius increase is mostly linear, with a short exponential window at the beginning, due to the fact that contact inhibition becomes more and more relevant over time in the core as cells proliferate.

I created line charts for both radius and tumor area change over time to inspect the effects. I also calculated front velocity at each execution, which is somewhat difficult to determine whether it is realistic or not, because reported values range from $\sim 0.01 \mu\text{m}/\text{min}$ to even above $0.15 \mu\text{m}/\text{min}$ [2] [5].

Based on a dataset of 50 simulations (each from a single STC, 12 hour growth) with default parameters (immune response included), we can conclude that the radius chart matches the expected exponential, then linear growth data (figure 3.1). Radial front speed is $0.01 \pm 0.004 \mu\text{m}/\text{min}$. This is around the lower limit of the previously described values, which could indicate a bit slower proliferation or migration than in most of the real scenarios. One parameter in particular that can be a cause for this is the chance of symmetric division (PS) of stem cells, which is by default at 25%, as in the code of the original model it is set to an even lower number of 10% at the time of writing [1], but is reported to be much higher in many tumors, even reaching above 80% in some cases [17].

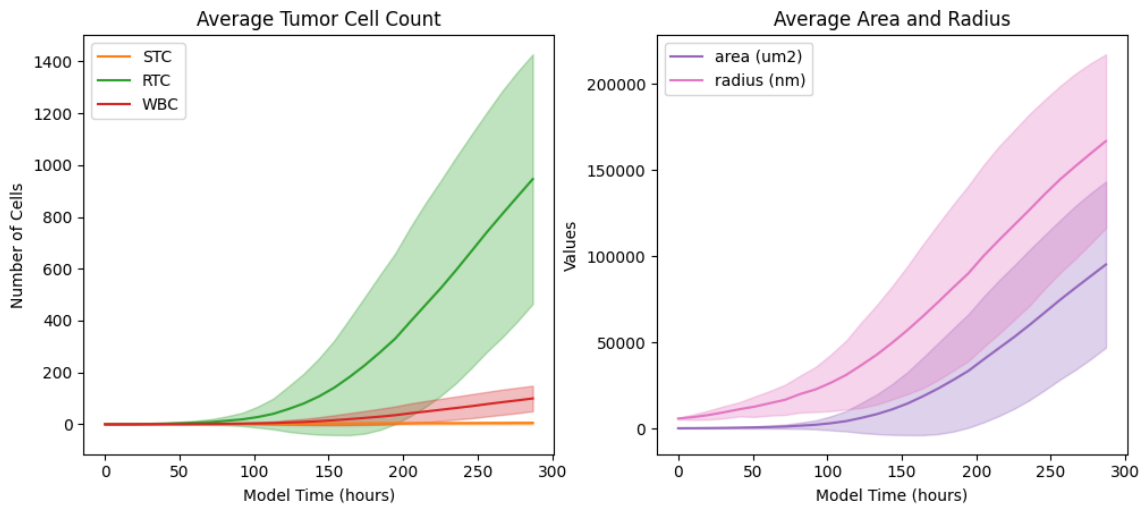


Figure 3.1: Growth plots of 50 simulations with default parameters over 12 hours.

3.2. The immune response

All simulations of the dataset worked with an $I = 3$ immune strength and the default parameters of the immune response function itself (described in the TRP work, along with all the equations for cell spawning, movement, immune exhaustion, immune-tumor interactions, etc.). The mean ratio of the immune and the cancer cells was 0.138 ± 0.07 , showing valid results, but a significant variability in the dataset. Animation of the growth confirmed the coded behaviour of the white blood cells to spawn around the tumor (in a "frame"), move towards colony density at a much higher speed (taking multiple steps per cycle), and upon contact with targets, kill them with a probability based on I . On the other hand, the kills/cell/day count was lower than expected: 1.553 ± 0.36 , indicating the need for a possible increase in kill probability (at $I = 3$ it is between 0.07 and 0.3 based on the mutation state of the cell).

3.3. Proliferation, contact inhibition

While the proliferation probability itself depends on the cell cycle time (CCT) and time step (dT) parameters, divisions can be inhibited by the lack of space on the field, meaning no free neighboring cells. Proliferation is constrained to the boundary during the growth, where more space is available. This property is simulated well by the model: a dense core quickly develops in most scenarios, and the animation clearly shows higher division activity at the edges.

3.4. Circularity of the border

When looking at the shape of the colony, it is expected to be roughly circular with (potentially very) irregular boundary. During a single simulation, a plot of borders at different hours is created and circularity is calculated from the area and perimeter ($C = 4\pi A/p^2$).

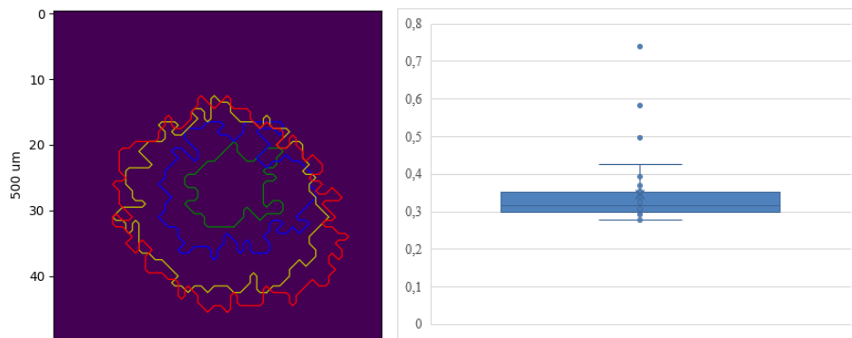


Figure 3.2: On the left, a plot of boundaries during the growth in a single simulation. On the right, a boxplot of mean circularity values of 50 simulations with default parameters.

The simulations most of the time result in colonies that resemble a circle, especially at the second half of the growth. The average circularity (determined from the mean circularity of 5 time points in each run, with default parameters) is 0.346 ± 0.09 , which means a pretty irregular boundary, as expected. In a cellular automaton model, due to pixelation and lattice-dependence, these values are artificially lower, even a "perfect circle" would not lead to $C = 1$ on a grid, so I would not call this number too low. While other articles reported higher ones (>0.7) [18], they did not study an immune response, which can strongly alter circularity. Without immune cells and increased cell adhesion, values above 0.5 can be achieved with the model, and considering its pixelated cellular automaton nature, we can say that it reflects rough tumors relatively well.

3.5. Model tuning

Based on the previously described results, I experimented with tuning some parameters of the model to achieve higher realism in the aspects that showed slight deviation from literature. Mainly, how many targets an immune cell kills per day (which was lower than reported) and radial front speed (which matched very slow tumors). Tuning up the kill chance on contact by a small amount, from 0.07-0.3 to 0.15-0.4 at $I = 3$ increased average kills/cell/day to 3.315 ± 0.67 , without a strong impact on tumor growth behavior, leading to a small drop in radial front speed.

To counter this effect and increase radial growth, I experimented with higher PS parameter values. However, this did not really lead to faster fronts in the simulations, even though the ratio of STCs and RTCs became larger. Even with a migration capacity (μ) of 10 (2.5 times the default) we cannot see a significant rise in the average radial front speed. There are many individual runs where this value exceeds 0.015 and reaches 0.2, and other times this gets extremely low due to the immune system overtaking. The low average result can be partially due to this effect: the immune system winning over the cancer. Other reasons can include contact inhibition caused by the immune cells, as in this model a tumor cell and a white blood cell cannot exist on the same grid spot, so the latter can block divisions. (This approach could be revisited later, since WBCs are very small and can even move between tissue.) Moreover, a cellular automaton model uses discrete times on a grid with only $10 \mu\text{m}$ jumps outwards, which is a pretty small cell diameter. Overall, the simulated front speed is still realistic, but mostly for slower tumors.

4. Summary

4.1. Validation results

In this work, I detailed the results of the validation of my tumor cellular automaton model. I analyzed several articles about real cancer (and immune system) behavior and extracted important metrics, such as the shape of growth curves, front speed, border irregularity, etc. I run a large number of simulations to identify parameter influence and determine average growth values for comparison.

Many properties of the resulting colonies were realistic: linear radius growth, proliferation constrained to the border, roughly circular shape, and good values related to the immune system (ratio of cell types and kills per day after some tuning, etc.). Some aspects were just in the appropriate ranges, like radial front speed at around the minimum observed values or the circularity (boundary irregularity) measure. Considering the cellular automaton nature of the model, and the presence of immune cells not part of the experiments reported in most papers, the simulation results are encouraging, and potentially optimizable even further.

4.2. Future development

In the future, more aspects of the model need to be reviewed, such as how mutations work (since they have a very primitive implementation). More data needs to be collected from publications to improve this system, and more options/customization must be added for the machine learning part. The TCAMpy module and the dashboard will require consistent updates to reflect the upcoming changes well. Knowing that the current version shows realistic tumor behavior, steps can be taken to extend these features and develop new ones based on previous experimental findings.

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